Introduction to Probability and Statistics

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Statistics for Engineers



Course Code: STAT 251
Course Title: Introduction to Probability and Statistics
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Name of the Topics of this Course

- Statistics and Data Description
- Probability
- Probability Distribution
- Stochastic Process and Queueing Theory



Statistics

Seligman defines **Statistics** as the science which deals with the methods of collecting classifying, presenting, comparing and interpreting numerical data collected to throw some light on any sphere of enquiry.

The Merriam-Webster's Collegiate Dictionary definition is: "A branch of mathematics dealing with the collection, analysis, interpretation, and presentation of masses of numerical data."

The American Heritage Dictionary defines statistics as: "The mathematics of the Collection, organization, and interpretation of numerical data, especially the analysis of population characteristics by inference from sampling."



Statistics and Data Description

A simple but concise definition by Croxton and Cowden: "Statistics is defined as the collection, presentation, analysis and interpretation of numerical data."

A comprehensive definition of statistics can be collection, organisation, presentation, analysis and interpretation of numerical data.

In other words it can be defined as - "Statistics is the science and the branch of applied mathematics which concerns with the collection of data, classifying and organization of data, presenting, comparing, finding the hidden information of data, interpreting and drawing inference about data."



Statistics and Data Description

Some important terms related to Statistics-

- Variable - various values of the characteristic recorded from different unit are expressed by quantity, the characteristic is called variable.
  - Qualitative variable
  - Quantitative variable
    - Discrete variable
    - Continuous variable

    The variable, Discrete variable or Continuous variable may again classified as Random variable (Discrete random variable and Continuous random variable) and Non-random variable.
- Population - total of units which are under investigation according to predetermined objective.
- Sample - representative number of population units.



Statistics and Data Description

- Data - any measurement of one or more variable(s) recorded either from population units or sample units is known as data.
  - Raw data - collected from sampling units and stored without any systematic fashion.
  - Classified data - data are presented in a systematic fashion in rows or columns or even in ordered way.
  - Primary data - collected from sampling units by personal interview or by mailed questionnaire.
  - Secondary data - collected from official record or from published work.



Statistics and Data Description

- Measurement and measurement scales of Data
  - Nominal Scale - the lowest measurement scale is the nominal scale. As the name implies it consists of "naming" observations or classifying them into various mutually exclusive and collectively exhaustive categories.
  - Ordinal Scale- whenever observations are not only different from category to category but can be ranked according to some criterion.
  - Interval Scale - the interval scale is a more sophisticated scale than the nominal or ordinal in that with this scale not only is it possible to order measurements, but also the distance between any two measurements is known
  - Ratio Scale - the highest level of measurement is the ratio scale. This scale is characterized by the fact that equality of ratios as well as equality of intervals may be determined. Fundamental to the ratio scale is a true zero point. The measurement of such familiar traits as height, weight, and length makes use of the ratio scale.



Statistics and Data Description

Let $x_1, x_2, ..., x_n$ be the random sample from a population and $x_{(1)}, x_{(2)}, ..., x_{(n)}$ be the ordered sample. Then,

Total $= x_1 + x_2 + ... + x_n = \sum_{i=1}^{n} x_i$

Product $= x_1 \times x_2 \times ... \times x_n = \prod_{i=1}^{n} x_i$

Range, $R = Maximum - Minimum = x_{(n)} - x_{(1)}$

Sturge's Formula for number of class, $k = 1 + 3.332 \log_{10}(n)$

Class width, $h = \dfrac{R}{k}$

Arithmetic Mean, AM $= \bar{x} = \dfrac{1}{n} \sum_{i=1}^{n} x_i$

Geometric Mean, GM $= (x_1 \times x_2 \times ... \times x_n)^{\frac{1}{n}} = \left(\prod_{i=1}^{n} x_i \right)^{\frac{1}{n}}$

$$GM = \text{Anti log} \frac{1}{n} \sum_{i=1}^{n} \log_{10}(x_i) = 10^{\left[\frac{1}{n} \sum_{i=1}^{n} \log_{10}(x_i) \right]}$$



$$\text{Harmonic Mean, HM} = \frac{n}{\frac{1}{x_1} + \frac{1}{x_2} + \ldots + \frac{1}{x_n}} = \frac{n}{\sum_{i=1}^{n} \frac{1}{x_i}}$$

Trimmed Mean

Trimmed mean is a compromise between $\bar{x}$ and $\tilde{x}$. A 10 percent trimmed mean, for example, would be computed by eliminating the smallest 10 percent and the largest 10 percent of the sample and then averaging what is left over.

$\tilde{x} = \bar{x}_{tr(\alpha)} =$ α percent trimmed mean.

Winsorized Mean

Let $x_1, x_2, \ldots, x_{10}$ be the random sample and $x_{(1)}, x_{(2)}, \ldots, x_{(10)}$ be the ordered sample. Then the 10% winsorized mean is defined as-

$$WM = \frac{x_{(2)} + x_{(2)} + x_{(3)} + x_{(4)} + x_{(5)} + x_{(6)} + x_{(7)} + x_{(8)} + x_{(9)} + x_{(9)}}{10}$$



Interquartile mean (IQM)

The interquartile mean (IQM) (or midmean) is a statistical measure of central tendency based on the truncated mean of the interquartile range. In calculation of the IQM, only the data between the first and third quartiles is used, and the lowest 25% and the highest 25% of the data are discarded.

Let $x_1, x_2, ..., x_n$ be the random sample and $x_{(1)}, x_{(2)}, ..., x_{(n)}$ be the ordered sample. Then, $x_{IQM} = \frac{2}{n} \sum_{i=(\frac{n}{4}+1)}^{\frac{3n}{4}} x_{(i)}$



Tukey's Trimean or Trimean (TM)

$TM = \frac{Q_1 + 2Q_2 + Q_3}{4}$ where Q_i be the ith quartile.

Quartile

Quartile is the measure of location in the array of data set which divides the array into four equal parts. There are three quartiles (i) first quartile Q_1 (ii) second quartile Q_2 (iii) third quartile Q_3. Let $x_1, x_2, ..., x_n$ be the random sample and $x_{(1)}, x_{(2)}, ..., x_{(n)}$ be the ordered sample. Then,

$$Q_i = x_{\left(\frac{i(n+1)}{4}\right)} \text{ if } n \text{ is odd}$$

$$Q_i = \frac{1}{2}\left[x_{\left(\frac{in}{4}\right)} + x_{\left(\frac{in}{4}+1\right)}\right] \text{if } n \text{ is even}$$

Inter Quartile Range, $IQR = Q_3 - Q_1$



Percentile

Percentile is the measure of location in the array of data set which divides the array into hundred equal parts. There are ninety nine percentiles.

$$P_i = x_{\left(\frac{i(n+1)}{100}\right)} \text{ if } n \text{ is odd}$$

$$P_i = \frac{1}{2}\left[x_{\left(\frac{in}{100}\right)} + x_{\left(\frac{in}{100}+1\right)}\right] \text{if } n \text{ is even}$$

Mode

Mode is the value of the variable which occurs most frequently in the series of the observation of the variable.



Median

Median is the value of the ordered series which divides the array into two equal parts such that half of observation bellow it and other half of observation are above it. Let $x_1, x_2, ..., x_n$ be the random sample and $x_{(1)}, x_{(2)}, ..., x_{(n)}$ be the ordered sample.

$$Me = x_{\left(\frac{(n+1)}{2}\right)} \text{ if } n \text{ is odd}$$

$$Me = \frac{1}{2}\left[x_{\left(\frac{n}{2}\right)} + x_{\left(\frac{n}{2}+1\right)}\right] \text{if } n \text{ is even}$$



Exercise-01

The data represents the weight (kg) of the thirty five students of the department of Computer Science and Engineering at Jahangirnagar University: 36, 37, 40, 41, 41, 44, 44, 45, 47, 48, 48, 50, 51, 51, 51, 51, 52, 55, 55, 55, 55, 57.5, 59, 60, 60, 64, 65, 67, 68, 70, 75, 75, 78, 78, and 87. Now include your weight (kg) with the above data then evaluate Mean, Median, Mode, Trimmed mean, Winsorized Mean, Interquartile mean, Quartiles, Inter Quartile Range, Trimean, and P_{60}.



Exercise-02

The data represents the weight (kg) of the thirty five students of the department of Computer Science and Engineering at Jahangirnagar University: 36, 37, 40, 41, 41, 44, 44, 45, 47, 48, 48, 50, 51, 51, 51, 51, 52, 55, 55, 55, 55, 57.5, 59, 60, 60, 64, 65, 67, 68, 70, 75, 75, 78, 78, and 87. Using suitable class interval construct a frequency table. Also calculate relative frequency, cumulative frequency, and cumulative relative frequency.

Solution

Here, Minimum=36, Maximum= 87, R=87-36=51, $n = 35$

$k = 1 + 3.332 \log_{10}(n) = 6.14 \simeq 7$ and $h = \frac{R}{k} = 8.30 \simeq 9$



Statistics and Data Description

Class Interval	Frequency	Cumulative Frequency	Relative Frequency	Cumulative Relative frequency
35-44	5	5	0.143	0.143
44-52	11	16	0.314	0.457
52-60	7	23	0.200	0.657
60-68	5	28	0.143	0.800
68-76	4	32	0.114	0.914
76-84	2	34	0.057	0.971
84-92	1	35	0.029	1.000



Stem-and-leaf plot

Stem-and-leaf plot is a tabular presentation of data where each data value is split into a "stem" or "the first digit or digits" and a "leaf" or "usually the last digit". Procedure to make stem-and-leaf plot is separate each observation or data into a stem which will consist of all except rightmost digit and leaf, the rightmost digit.

Example: Construct a stem-and-leaf display for the weight (kg) of students 36, 37, 40, 41, 41, 44, 44, 45, 47, 48, 48, 50, 51, 51, 51, 51, 52, 55, 55, 55, 55, 58, 59, 60, 60, 64, 65, 67, 68, 70, 75, 75,78, 78, and 87.

Stem	Leaf
3	6,7
4	0, 1, 1, 4, 4, 5, 7, 8, 8
5	0, 1, 1, 1, 1, 2, 5, 5, 5, 5, 8, 9
6	0, 0, 4, 5, 7, 8
7	0, 5, 5, 8, 8
8	7



Box-and-Whisker Plot

Example: Draw a Box-and-whisker plot for the following data- 0.1, 0.1, 0.4, 0.4, 0.5, 0.5, 0.6, 0.6, 0.6, 0.8, 0.8, 0.9, 0.9, 0.9, 0.9, 0.9, 0.9, 0.9, 0.9, 1.0, 1.0, 1.0, 1.1, 1.1, 1.1, 1.2, 1.2, 1.3, 1.3, 1.3, 1.3, 1.4, 1.4, 1.4, 1.5, 1.6, 1.6, 1.7,1.7,1.7, 1.8, 1.9, 1.9, 2,2, 2,2, 2.0, 2.1, 2.1, 2.3, 2.3, 2.4, 2.4, 2.5, 2.7, 2.7, 2.7, 2.8, 2.8, 2.9, 2.9, 3,3, 3.1, 3.1, 3.3, 3.3, 3.5, 3.8, 3.8, 3.9, 4.0, 4.1, 4.2, 4.2, 4.4, 4.4, 4.5, 4.6, 4.8, 4.8, 4.8, 4.9, 5.0, 5.1, 5.1, 5.2, 5.3, 5.5, 5.5, 5.8, 5.8, 6.2, 6.3, 6.7, 7.8, 8.1, 8.2,and 13.5.



Box-and-Whisker Plot

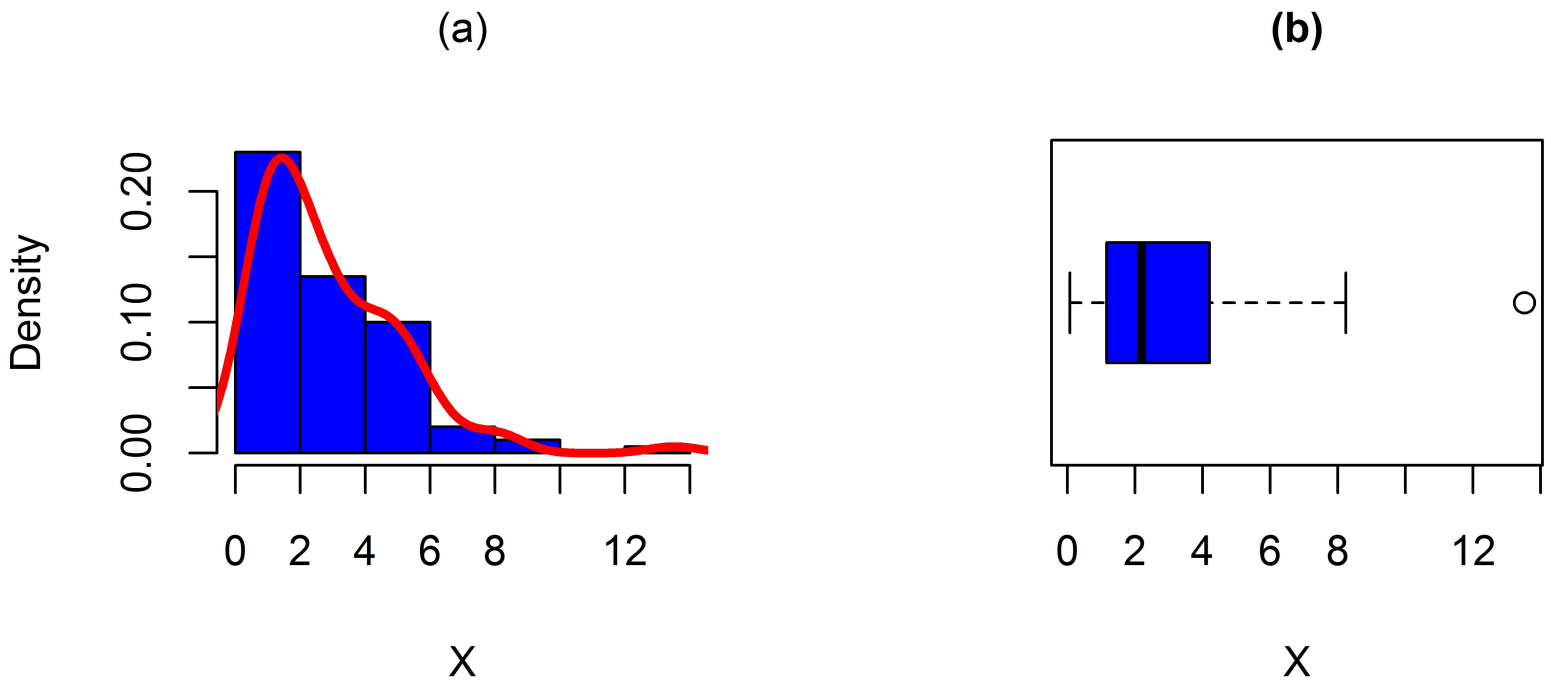


Figure: (a) Histogram with density and (b) Box-and-whisker Plot

Empirical relationship between Mean, Median, and Mode is presented by Karl Pearson is -

Mode=3 Median - 2 Mean

Variance and Standard Deviation

Variance measures the average amount of variation of observations.
Standard deviation measures the average distance of the observations from mean.

$$S_{xx} = SS(x) = \sum_{i=1}^{n}(x_i - \bar{x})^2 = \sum_{i=1}^{n} x_i^2 - n\bar{x}^2$$

$$S_{xy} = SP(xy) = \sum_{i=1}^{n}(x_i - \bar{x})(y_i - \bar{y}) = \sum_{i=1}^{n} x_i y_i - n\bar{x}\bar{y}$$



Statistics and Data Description

Population Mean: $\mu = \frac{1}{N}\sum_{i=1}^{N} x_i$

Population Variance: $\sigma_x^2 = \frac{1}{N}\sum_{i=1}^{N}(x_i - \mu)^2$

Population Standard Deviation: $\sigma_x = \sqrt{\frac{1}{N}\sum_{i=1}^{N}(x_i - \mu)^2}$

Population Covariance: $COV(xy) = \frac{1}{N}\sum_{i=1}^{N}(x_i - \mu_x)(y_i - \mu_y)$

Sample Mean: $\bar{x} = \frac{1}{n}\sum_{i=1}^{n} x_i$

Sample Variance: $s_x^2 = \frac{1}{n-1}\sum_{i=1}^{n}(x_i - \bar{x})^2$

Sample Standard Deviation: $s_x = \sqrt{\frac{1}{n-1}\sum_{i=1}^{n}(x_i - \bar{x})^2}$

Sample Covariance: $cov(xy) = \frac{1}{n-1}\sum_{i=1}^{n}(x_i - \bar{x})(y_i - \bar{y})$



Measure of Dispersion

The term dispersion means the scatteredness of observations from some central value. The term "measure of dispersion" is defined as an average deviation of observations from some central value. There are two types of measure of dispersion (i) Absolute measure of dispersion (ii) Relative measure of dispersion.

- Absolute measure of dispersion
  - Range, $R = x_{(n)} - x_{(1)}$
  - Mean Deviation

    $MD(Mean) = \frac{1}{n}\sum_{i=1}^{n}|x_i - \bar{x}|$

    $MD(Median) = \frac{1}{n}\sum_{i=1}^{n}|x_i - Me|$

    $MD(Mode) = \frac{1}{n}\sum_{i=1}^{n}|x_i - Mo|$
  - Standard deviation, $s_x = \sqrt{\frac{1}{n-1}\sum_{i=1}^{n}(x_i - \bar{x})^2}$
  - Quartile deviation, $QD = \frac{Q_3 - Q_1}{2}$



- Relative measure of dispersion
  - Coefficient of Range $= \dfrac{x_{(n)} - x_{(1)}}{x_{(n)} + x_{(1)}} \times 100\%$
  - Coefficient of Mean Deviation

    Coefficient of MD(mean) $= \dfrac{MD(mean)}{\bar{x}} \times 100\%$

    Coefficient of MD(median) $= \dfrac{MD(median)}{Median} \times 100\%$

    Coefficient of MD(mode) $= \dfrac{MD(mode)}{Mode} \times 100\%$
  - Coefficient of Variation, $CV = \dfrac{s_x}{\bar{x}} \times 100\%$
  - Coefficient of Quartile Deviation, $\dfrac{Q_3 - Q_1}{Q_3 + Q_1} \times 100\%$

Coefficient of Variation

A measure of relative variation called the sample coefficient of variation. It is defined as - $CV = \dfrac{s_x}{\bar{x}} \times 100\%$



Shape Characteristics of Distribution

The important measure of dispersion is variance which is known as mean square deviation or mean deviation of second order(or degree). The mean deviation of $r-th$ order can be calculated and which is known as $r-th$ central moment.

$$\mu_r = \frac{1}{N}\sum_{i=1}^{N}(x_i - \mu)^r$$

Skewness and Kurtosis

The asymmetric property of the curve is known as Skewness and the height characteristic of frequency curve is known as Kurtosis.



Shape Characteristics of Distribution

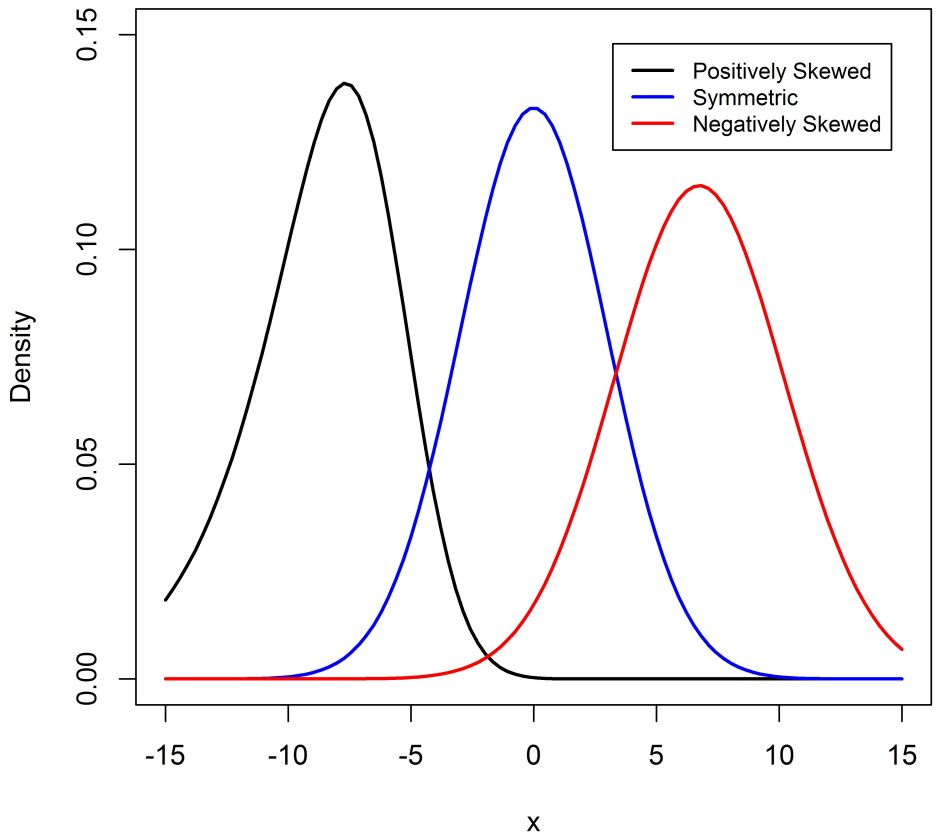


Figure: Skewness

Shape Characteristics of Distribution

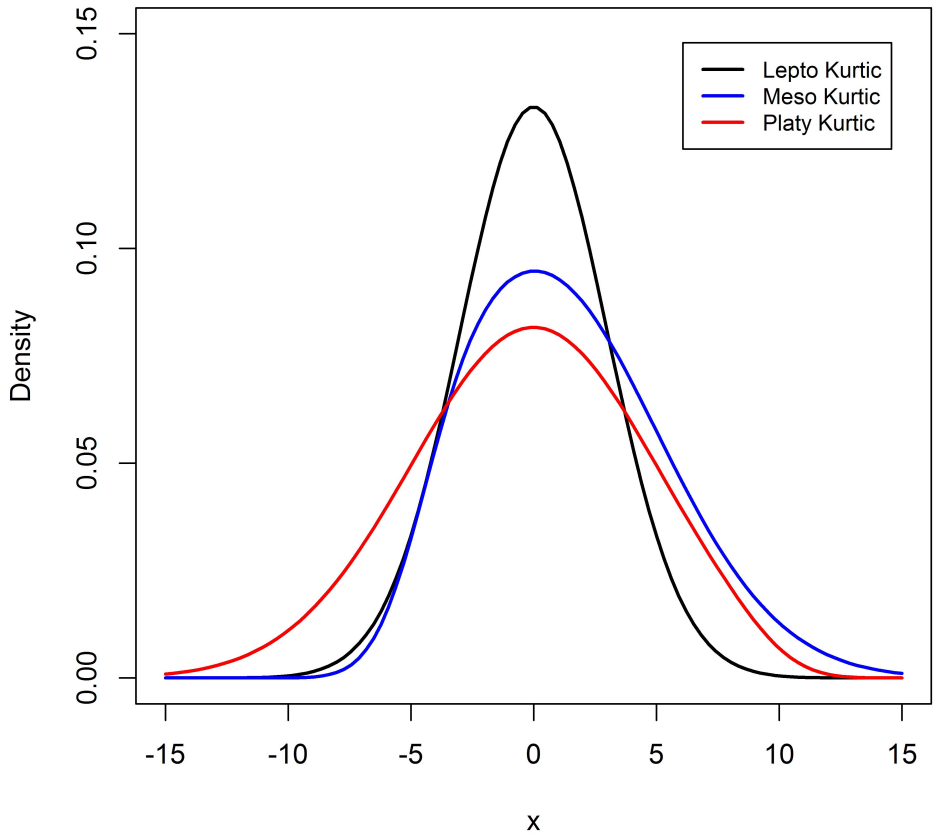


Figure: Kurtosis

Statistics and Data Description

Karl Pearson's Coefficient of Skewness, $\beta_1 = \frac{\mu_3^2}{\mu_2^3}$ and $\gamma_1 = \sqrt{\beta_1} = \frac{\mu_3}{\sigma^3}$.

If $\gamma_1 < 0$ and $\mu_3 < 0$ then the distribution is negatively skewed.
If $\gamma_1 = 0$ and $\mu_3 = 0$ then the distribution is symmetric.
If $\gamma_1 > 0$ and $\mu_3 > 0$ then the distribution is positively skewed.

Karl Pearson's Coefficient of Kurtosis, $\beta_2 = \frac{\mu_4}{\mu_2^2}$ and $\gamma_2 = \beta_2 - 3 = \frac{\mu_4}{\sigma^4} - 3$.

If $\gamma_2 > 0$ and $\beta_2 > 3$ then the distribution is Lepto kurtic.
If $\gamma_2 = 0$ and $\beta_2 = 3$ then the distribution is Meso kurtic.
If $\gamma_2 < 0$ and $\beta_2 < 3$ then the distribution is Platy kurtic.

If the data represent measurements on all the units of a finite population or a census, then these equations of Karl Pearson's Coefficient of Skewness and kurtosis should be utilized directly to determine these measures.



Statistics and Data Description

If the data being analyzed represent measurement of a variable made on sample units, the sample estimate of γ_1 and γ_2 are-

Skewness: $\gamma_1 = \frac{n}{(n-1)(n-2)} \sum_{i=1}^{n} \left(\frac{x_i - \bar{x}}{s_x} \right)^3$

Kurtosis: $\gamma_2 = \frac{n(n+1)}{(n-1)(n-2)(n-3)} \sum_{i=1}^{n} \left(\frac{x_i - \bar{x}}{s_x} \right)^4 - \frac{3(n-1)^2}{(n-2)(n-3)}$

These values also calculated from Excel. https://support.microsoft.com/en-us/office/skew-function-bdf49d86-b1ef-4804-a046-28eaea

https://support.microsoft.com/en-us/office/kurt-function-bc3a



Grouped Data: Frequency Table

Class Interval	Mid value, x_i	Frequency, f_i	Cumulative Frequency
$x-[x+1h]$	x_1	f_1	f_1
$[x+1h]-[x+2h]$	x_2	f_2	f_1+f_2
$[x+2h]-[x+3h]$	x_3	f_3	$f_1+f_2+f_3$
.	.	.	.
.	.	.	.
$[x+(k-1)h]-[x+kh]$	x_k	f_k	$\sum_{i=1}^{k} f_i$
Total		$\sum_{i=1}^{k} f_i = n$	



0



Mean and median for grouped data

Arithmetic Mean, $AM = \bar{x} = \frac{1}{n}\sum_{i=1}^{k} f_i x_i$

Geometric Mean, $GM = \left(x_1^{f_1} x_2^{f_2} \ldots x_k^{f_k}\right)^{\frac{1}{n}} = 10^{\left[\frac{1}{n}\sum_{i=1}^{k} f_i \log(x_i)\right]}$

Harmonic Mean, $HM = \frac{n}{\sum_{i=1}^{k} \frac{f_i}{x_i}}$

Median, $Me = L + \frac{h}{f}\left[\frac{n}{2} - c\right]$

Median is the value falling in a class for which cumulative frequency is just greater than or equal to $\frac{n}{2}$. Here, L = Lower limit of the median class, h = width of the median class, f = frequency of the median class, c = cumulative of the class preceding of the median class, and n = total frequency.



Mode for grouped data

$$\text{Mode, } Mo = L + \frac{h(f_m - f_p)}{(f_m - f_p) + (f_m - f_a)}$$

The class which having the maximum frequency in a frequency distribution is known as Modal class.

Here,
L = lower limit of the modal class
h = width of the modal class
f_p = frequency of the class preceding the modal class
f_m = frequency of the modal class
f_a = frequency of the class following the modal class



Quartile for grouped data

Quartile, $Q_i = L + \frac{h}{f}\left[\frac{in}{4} - c\right]$, $i = 1, 2, 3$.

Q_i class is that class for which cumulative frequency is just greater than or equal to $\frac{in}{4}$.

Here,
L = Lower limit of the Q_i class
h = width of the Q_i class
f = frequency of the Q_i class
c = cumulative of the class preceding of the Q_i class
n = total frequency



Variance, Standard deviation, central moment for grouped data

Sample Variance, $Var(x) = s^2 = \frac{1}{n-1}\sum_{i=1}^{k} f_i(x_i - \bar{x})^2$

$$s^2 = \frac{1}{n-1}\left[\sum_{i=1}^{k} f_i x_i^2 - \frac{(\sum_{i=1}^{k} f_i x_i)^2}{n}\right]$$

Sample Standard deviation, $s = \sqrt{Var(x)} = \sqrt{s^2}$

Central moment, $\mu_r = \frac{1}{N}\sum_{i=1}^{k} f_i(x_i - \bar{x})^r,\ r = 0, 1, 2, 3, 4, ...$

$\mu_o = 1, \mu_1 = 0, \mu_2 = \sigma^2$



Exercise-02

The data represents the weight (kg) of the thirty five students of the department of Computer Science and Engineering at Jahangirnagar University: 36, 37, 40, 41, 41, 44, 44, 45, 47, 48, 48, 50, 51, 51, 51, 51, 52, 55, 55, 55, 55, 57.5, 59, 60, 60, 64, 65, 67, 68, 70, 75, 75, 78, 78, and 87. Now include your weight (kg) with the above data then evaluate Mean, Median, Mode, Trimmed mean, Winsorized Mean, Interquartile mean, Quartiles, Inter Quartile Range, Trimean, and P_{60}.

Solution Home Work.



Exercise-03

The data represents the weight (kg) of the thirty five students of the department of Computer Science and Engineering at Jahangirnagar University: 36, 37, 40, 41, 41, 44, 44, 45, 47, 48, 48, 50, 51, 51, 51, 51, 52, 55, 55, 55, 55, 57.5, 59, 60, 60, 64, 65, 67, 68, 70, 75, 75, 78, 78, and 87. Using suitable class interval construct a frequency table. Also calculate mean, median, mode, third quartile, standard deviation, skewness, and kurtosis.

Solution

Here, Minimum=36, Maximum= 87, R=87-36=51, $n = 35$

$k = 1 + 3.332 \log_{10}(n) = 6.14 \simeq 7$ and $h = \frac{R}{k} = 8.30 \simeq 9$



Solution of Exercise-03....

CI	x_i	f_i	c.f	$f_i x_i$	$f_i(x_i - \bar{x})^2$	$f_i(x_i - \bar{x})^3$	$f_i(x_i - \bar{x})^4$
35-44	39.5	5	5	197.5	1428.1	-24134	407865.4
44-52	48	11	16	528.0	776.2	-6519.7	54765.8
52-60	56	7	23	392.0	1.1	-0.4	0.2
60-68	64	5	28	320.0	288.8	2194.9	16681.1
68-76	72	4	32	288.0	973.4	15185.7	236896.4
76-84	80	2	34	160.0	1113.9	26288.5	620408.9
84-92	88	1	35	88.0	998.6	31554.5	997122.1
Total		35		1973.5	5580.1	44569.5	2333739.9

$$Mean = \bar{x} = \frac{1}{n}\sum_{i=1}^{k} f_i x_i = \frac{1}{35} \times 1973.5 = 56.4$$



Solution of Exercise-03....

Median

Here, Median class: 52-60, L=52, h=9, f=7, n=35, c=16

$$Me = L + \frac{h}{f}\left[\frac{n}{2} - c\right] = 52 + \frac{9}{7}\left[\frac{35}{2} - 16\right] = 53.9$$

Mode

Here, Modal Class: 44-52, L=44, h=9, $f_p = 5$, $f_m = 11$, $f_a = 7$

$$Mo = L + \frac{h(f_m - f_p)}{(f_m - f_p) + (f_m - f_a)} = 44 + \frac{9(11-5)}{(11-5)+(11-7)} = 49.5$$

Since $[Mean = 56.4] > [Median = 53.9] > [Mode = 49.5]$, the distribution of the data is positively skewed.



Solution of Exercise-03....

Third Quartile

Here, i=3, Third Quartile class: 60-68, L=60, h=9, f=5, n=35, c=23

$$Q_3 = L + \frac{h}{f}\left[\frac{3n}{4} - c\right] = 60 + \frac{9}{5}\left[\frac{3 \times 35}{4} - 23\right] = 65.85$$

Standard Deviation

Sample Variance, $Var(x) = s^2 = \frac{1}{n-1}\sum_{i=1}^{k} f_i(x_i - \bar{x})^2 = \frac{5580.1}{35-1} = 164.1$

Sample Standard deviation, $s = \sqrt{Var(x)} = \sqrt{s^2} = \sqrt{164.1} = 12.8$



Solution of Exercise-03....

Skewness and Kurtosis

Let assume that the sample mean is the unbiased estimate of population mean. Here, $\mu_2 = \frac{1}{35}\sum_{i=1}^{k=7} f_i(x_i - \bar{x})^2 = \frac{1}{35} \times 5580.1 = 164.1$

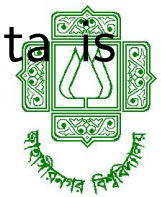
$\mu_3 = \frac{1}{35}\sum_{i=1}^{k=7} f_i(x_i - \bar{x})^3 = \frac{1}{35} \times 44569.5 = 1273.4$

$\mu_4 = \frac{1}{35}\sum_{i=1}^{k=7} f_i(x_i - \bar{x})^4 = \frac{1}{35} \times 2333739.9 = 66678.2$

Skewess: $\beta_1 = \frac{\mu_3^2}{\mu_2^3} = \frac{1273.4^2}{164.1^3} = 0.37$ and $\gamma_1 = \sqrt{\beta_1} = \frac{\mu_3}{\sigma^3} = 0.61 > 0$

Kurtosis: $\beta_2 = \frac{\mu_4}{\mu_2^2} = \frac{66678.2}{164.1^2} = 2.5$ and $\gamma_2 = \beta_2 - 3 = 2.5 - 3 = -0.5 < 0.$

Since $\gamma_1 = 0.61 > 0$ and $\gamma_2 = -0.5 < 0$ the distribution of the data is positively skewed and platy kurtic.



Correlation and Regression

Regression

Regression analysis is the technique to study the cause and effect relationship of two or more variables when variables are recorded from each sample point of a given sample.

Let (x_1, y_1), (x_2, y_2), . . ., (x_n, y_n) be the n pairs of samples for the random variable X and Y. Assume that X and Y follow the regression model-

$$Y = \beta_0 + \beta_1 X + \epsilon$$

The parameter β_1 is called the regression coefficient of Y on X which measures the rate of change of Y for a unit change in the value of X, β_0 is the intercept term which is the parameter measuring the value of Y in absence of X, and ϵ is called the error term.



Regression

Assumptions for Regression Analysis

- Dependent variable assumed to be normally distributed.
- Explanatory or independent variable(s) is/are non-random variable.
- Independent variables are uncorrelated.
- The error term is normally and independently distributed with $E(\epsilon_i) = 0$, $E(\epsilon_i, \epsilon_j) = 0$ for $i \neq j$ and $V(\epsilon_i) = \sigma^2$.



Correlation and Regression

Least Squares Estimation: Fitting a straight-line regression model
Let (x_1, y_1), (x_2, y_2), . . ., (x_n, y_n) be the n pairs of samples for the random variable X and Y. Assume that X and Y follow the regression model-

$$Y = \beta_0 + \beta_1 X + \epsilon$$

Assume that fitted regression is $\hat{Y} = \hat{\beta}_0 + \hat{\beta}_1 X$. Thus $\epsilon = Y - \hat{Y}$
The regression analysis indicate that the value of $\hat{\beta}_0$ and $\hat{\beta}_1$ are to be found out using n pairs of values in such a way that sum of square of deviations of Y and $\hat{Y}$ is minimum.

$$\sum_{i=1}^{n} \epsilon_i^2 = \sum_{i=1}^{n} (Y_i - \hat{Y}_i)^2$$
$$\sum_{i=1}^{n} \epsilon_i^2 = \sum_{i=1}^{n} (Y_i - \hat{\beta}_0 - \hat{\beta}_1 X_i)^2$$



Least Squares Estimation .....

Let, $f(\hat{\beta}_0, \hat{\beta}_1) = \sum_{i=1}^{n} \epsilon_i^2 = \sum_{i=1}^{n} (Y_i - \hat{\beta}_0 - \hat{\beta}_1 X_i)^2$

$f(\hat{\beta}_0, \hat{\beta}_1) = \sum_{i=1}^{n} (Y_i - \hat{\beta}_0 - \hat{\beta}_1 X_i)^2$

The value of $f(\hat{\beta}_0, \hat{\beta}_1)$ will be minimum if the values of $\hat{\beta}_0$ and $\hat{\beta}_1$ are found out in such a way that $\frac{\partial f}{\partial \hat{\beta}_0} = 0$ and $\frac{\partial f}{\partial \hat{\beta}_1} = 0$. These two equation gives

$\sum_{i=1}^{n} Y_i = n\hat{\beta}_0 + \hat{\beta}_1 \sum_{i=1}^{n} X_i$

$\sum_{i=1}^{n} X_i Y_i = \hat{\beta}_0 \sum_{i=1}^{n} X_i + \hat{\beta}_1 \sum_{i=1}^{n} X_i^2$

These equations are known as normal equations. After solving these two equations, we have-



Least Squares Estimation .....

$\hat{\beta}_0 = \bar{Y} - \hat{\beta}_1 \bar{X}$ and

$$\hat{\beta}_1 = \frac{\sum_{i=1}^{n} X_i Y_i - \frac{\sum_{i=1}^{n} X_i \sum_{i=1}^{n} Y_i}{n}}{\sum_{i=1}^{n} X_i^2 - \frac{(\sum_{i=1}^{n} X_i)^2}{n}} = \frac{SP(XY)}{SS(X)} = \frac{\frac{SP(XY)}{n-1}}{\frac{SS(X)}{n-1}} = \frac{Cov(XY)}{Var(X)}$$

$$SS(X) = \sum_{i=1}^{n} (X_i - \bar{X})^2 = \sum_{i=1}^{n} X_i^2 - n\bar{X}^2$$

$$SP(XY) = \sum_{i=1}^{n} (X_i - \bar{X})(Y_i - \bar{Y}) = \sum_{i=1}^{n} X_i Y_i - n\bar{X}\bar{Y}$$

Sample Mean: $\bar{X} = \frac{1}{n} \sum_{i=1}^{n} X_i$

Sample Variance: $s_X^2 = \frac{1}{n-1} \sum_{i=1}^{n} (X_i - \bar{X})^2$

Sample Covariance: $Cov(XY) = \frac{1}{n-1} \sum_{i=1}^{n} (X_i - \bar{X})(Y_i - \bar{Y})$



Correlation

The process of measuring the amount of linear relationship of two variables is known as correlation analysis. The amount of linear relationship is known as correlation coefficient. The value of correlation coefficient lies between -1 to 1. The sample correlation is defined as -

$$r_{XY} = \frac{Cov(X,Y)}{\sqrt{Var(X)Var(Y)}} = \frac{SP(XY)}{\sqrt{SS(X)SS(Y)}}$$

where,

$$SS(X) = \sum_{i=1}^{n}(X_i - \bar{X})^2 = \sum_{i=1}^{n} X_i^2 - n\bar{X}^2$$

$$SS(Y) = \sum_{i=1}^{n}(Y_i - \bar{Y})^2 = \sum_{i=1}^{n} Y_i^2 - n\bar{Y}^2$$

$$SP(XY) = \sum_{i=1}^{n}(X_i - \bar{X})(Y_i - \bar{Y}) = \sum_{i=1}^{n} X_i Y_i - n\bar{X}\bar{Y}$$



Statistics and Data Description

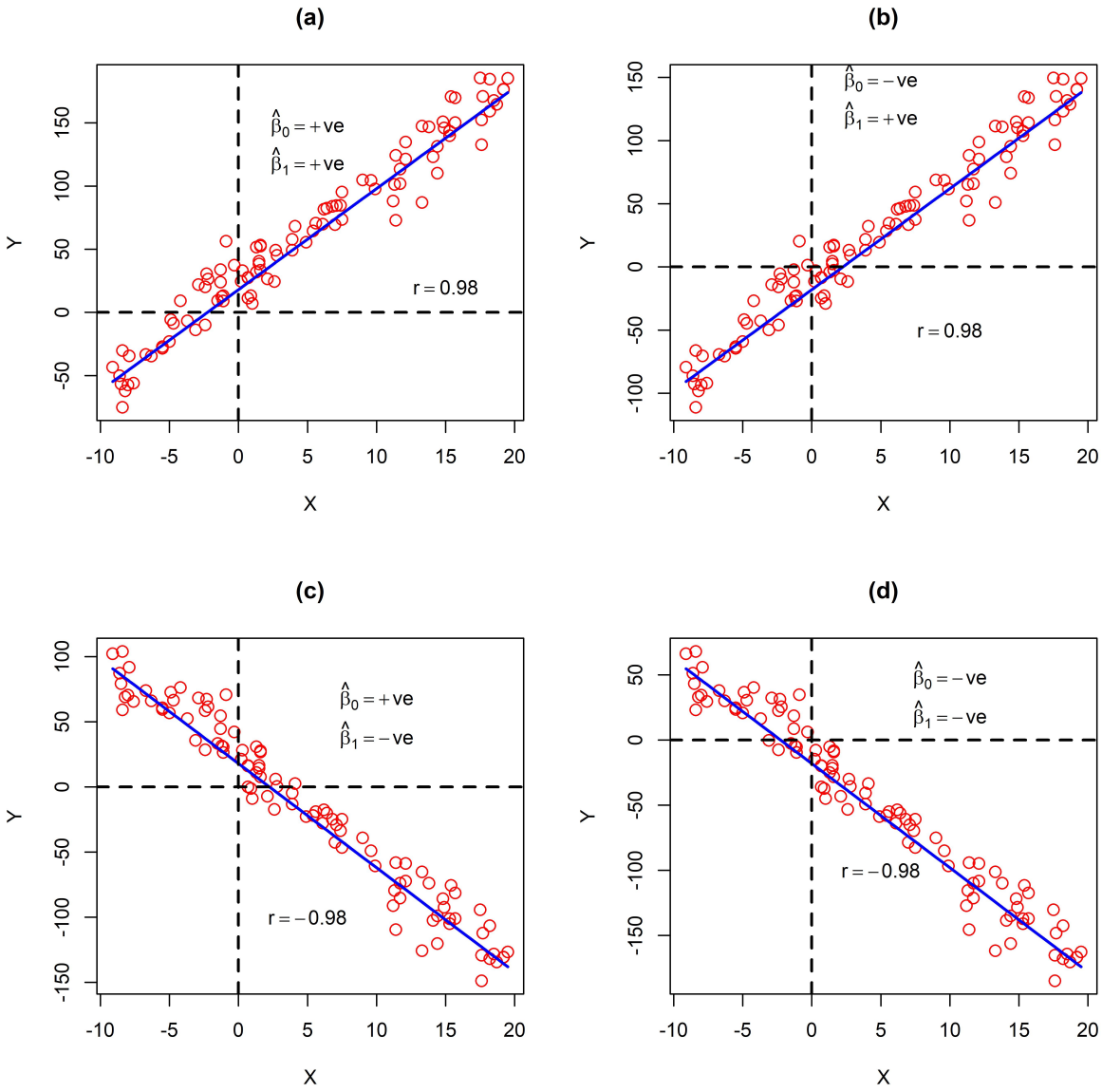


Figure: Schematic line plot for the variable X and Y and for (a) $\hat{\beta}_0 = 18$ and $\hat{\beta}_1 = 8$, (b) $\hat{\beta}_0 = -18$ and $\hat{\beta}_1 = 8$, (c) $\hat{\beta}_0 = 18$ and $\hat{\beta}_1 = -8$, (d) $\hat{\beta}_0 = -18$ and $\hat{\beta}_1 = -8$

If $\hat{Y} = \hat{\beta}_{0(yx)} + \hat{\beta}_{1(yx)}X$ and $\hat{X} = \hat{\beta}_{0(xy)} + \hat{\beta}_{1(xy)}Y$, then

(i) $r_{xy} = \sqrt{\hat{\beta}_{1(yx)}\hat{\beta}_{1(xy)}}$.

(ii) $r_{xy} = -\sqrt{\hat{\beta}_{1(yx)}\hat{\beta}_{1(xy)}}$ when $SP(xy)$ is negative.

(iii) $-1 \leq r_{xy} \leq 1$ and $0 \leq r^2 \leq 1$.

(iv) If $\hat{\beta}_{1(xy)} > 0$ then $\hat{\beta}_{1(yx)} < 0$.

The coefficient of determination, denoted by R^2 or r^2, is the proportion of the variance in the dependent variable that is predictable from the independent variable(s). Also, coefficient of determination is sometimes referred to as the "goodness of fit".



Positive Correlation: If X and Y have a strong positive linear correlation, r is close to $+1$. An r value of exactly $+1$ indicates a perfect positive fit. Positive values indicate a relationship between X and Y variables such that as values for X increase, values for Y also increase.

Negative Correlation: If X and Y have a strong negative linear correlation, r is close to -1. An r value of exactly -1 indicates a perfect negative fit. Negative values indicate a relationship between X and Y such that as values for X increase, values for Y decrease.

Spurious Correlation: If there is no linear correlation or a weak linear correlation, r is close to 0. A value near zero means that there is a random, non-linear relationship between the two variables.



Evans (1996) suggests for the absolute value of r:

$\lvert r \rvert$	Strength
0.00 - 0.19	Very weak correlation
0.20 - 0.39	Weak correlation
0.40 - 0.59	Moderate correlation
0.60 - 0.79	Strong correlation
0.80 - 1.00	Very strong correlation

Reference

Evans, J.D., 1996. Straightforward statistics for the behavioral sciences. Thomson Brooks/Cole Publishing Co.



Example: The data represents the height (in inch) and weight (in kg) for the twelve students. Assume that the data are normally distributed.

X(Height):	62	62	73	70	79	66	58	81	77	57	83	80
Y(Weight):	71	68	68	61	79	63	61	76	67	66	73	71

i) Fit a regression line to estimate weight.

ii) What is the value of $\hat{\beta}_0$ and $\hat{\beta}_1$?

iii) Evaluate the value of correlation coefficient's and coefficient's of determination.

iv) What is the value of weight for the value of height 72 inch?



Solution Here $\bar{X} = 70.7,\ \bar{Y} = 68.7$

X	Y	X^2	Y^2	XY
62	71	3844	5041	4402
62	68	3844	4624	4216
73	68	5329	4624	4964
70	61	4900	3721	4270
79	79	6241	6241	6241
66	63	4356	3969	4158
58	61	3364	3721	3538
81	76	6561	5776	6156
77	67	5929	4489	5159
57	66	3249	4356	3762
83	73	6889	5329	6059
80	71	6400	5041	5680
$\sum X_i = 848$	$\sum Y_i = 824$	$\sum X_i^2 = 60906$	$\sum Y_i^2 = 56932$	$\sum X_i Y_i = 58605$



Solution ...

(i) $\hat{\beta}_1 = \dfrac{SP(XY)}{SS(X)} = \dfrac{\sum_{i=1}^{n} X_i Y_i - n\bar{X}\bar{Y}}{\sum_{i=1}^{n} X_i^2 - n\bar{X}^2} = \dfrac{58605 - 12 \times 70.7 \times 68.7}{60906 - 12 \times 70.7^2} = 0.35$

$\hat{\beta}_0 = \bar{Y} - \hat{\beta}_1 \bar{X} = 68.7 - 0.35 \times 70.7 = 44$

$\hat{Y} = 44 + 0.35X$

That is, Weight $= 44 + 0.35$ Height

(ii) $\hat{\beta}_0 = 44$ and $\hat{\beta}_1 = 0.35$.

(iii) Here, $SS(X) = \sum_{i=1}^{n} X_i^2 - n\bar{X}^2 = 60906 - 12 \times 70.7^2 = 924.1$

$SS(Y) = \sum_{i=1}^{n} Y_i^2 - n\bar{Y}^2 = 56932 - 12 \times 68.7 = 295.7$

$SP(XY) = \sum_{i=1}^{n} X_i Y_i - n\bar{X}\bar{Y} = 58605 - 12 \times 70.7 \times 68.7 = 319.9$

Correlation, $r_{XY} = \dfrac{Cov(X,Y)}{\sqrt{Var(X)Var(Y)}} = \dfrac{SP(XY)}{\sqrt{SS(X)SS(Y)}} = \dfrac{319.9}{\sqrt{924.1 \times 295.7}} = 0.61$

Coefficient of determination, $r_{XY}^2 = 0.37$.

(iv) We have, Weight $= 44 + 0.35 \times$ Height $= 44 + 0.35 \times 72 = 69.2$



Statistics and Data Description



Statistics and Data Description



Python Code

```
# =====Read Data File =========#
import math
import numbers
import random
import scipy
import scipy.stats
import numpy as np # numerical computing
# data processing, CSV file I/O (e.g. pd.read_csv)
import pandas as pd
import matplotlib.pyplot as plt #visualization
import seaborn as sn #modern visualization
import statsmodels as sm
import statsmodels.api as sma
from statsmodels.sandbox.regression.predstd -
import wls_prediction_std

import statistics as st
from sklearn.linear_model import LinearRegression as L
```



Python Code

```
# =====Read Data File ==========#
dat=pd.read_csv("F:\Village of Study\
CSE - 251 INTRODUICTION TO PROBABILITY
AND STATISTICS\INFO_OF_CSE.csv")

dat.shape  # No of Row and Column of Data
# =====Write Data File ==========#
# dat.to_csv('D:\dat2.csv')

dat.head()   # To See First Five Rows
dat.tail()   # To See Last Five Rows

D2=dat[["Gender","Board_SSC","SSC_GPA",
  "Board_HSC","HSC_GPA","Weight","Height","Age"]]

DAT = pd.DataFrame(D2)
```



Python Code

```
DAT.count()
DAT.max()
DAT.min()
DAT.mean()
DAT.sem()   #Standard error of the mean
DAT.median()
DAT.mad() #Mean absolute deviation
DAT.var()
DAT.std()
DAT.skew()
DAT.kurt()
DAT.quantile(0.5)

DAT.describe()
DAT.describe(percentiles=[.05, .25, .75, .95])
```



Python Code

```
G=DAT[["Gender"]]
H=DAT[['Height']]
W=DAT[['Weight']]
A=DAT[['Age']]
H.describe()
#== Pairwise Scatter Plot =====#
plt.style.use('seaborn-white')
sn.pairplot(DAT[["Weight","Height","Age"]])
plt.show()

#====Pie Chart===#
from collections import Counter
print(Counter(DAT["Gender"]))
Counter(DAT["Gender"])
plt.style.use('seaborn-white')
plt.pie(np.array([20, 15]),
    labels = ["Male", "Female"], colors=["m","b"])
plt.title("Chart for Gender")
plt.legend(title = "Gender:")
plt.show()
```



Python Code

```
# Usual Boxplot
plt.style.use('seaborn-white')
sn.boxplot( y="Weight", color="b",data=DAT)
 # Add jitter with the swarmplot function.
sn.swarmplot(y='Weight', data=DAT, color="r")

#===line plot ====#
plt.style.use('seaborn-white')
plt.plot(DAT[['Weight']],marker = 'o', color="b",data=DAT)
plt.plot(DAT[['Height']],marker = 'o', color="r",data=DAT)
plt.plot(DAT[['Height']], DAT[['Weight']],marker = 'o', color="m")

##=======Scatter Plot ===#
plt.style.use('seaborn-white')
plt.scatter(DAT[['Height']],
  DAT[['Weight']],color="r",data=DAT)
plt.title("Plot for Height and Weight")
plt.xlabel("Height")
plt.ylabel("Weight")
plt.show()
```



Python Code

```
#====Histogram===#
plt.style.use('seaborn-white')
plt.hist(H,bins = 'auto',color="r")
plt.style.use('seaborn-white')
plt.hist(H, bins = 'sturges',normed=1,rwidth=0.95,color='r',alpha=0
```

```
#====Histogram===#
#plt.savefig("D:/NORMAl_HIST.jpg", dpi=500)
plt.style.use('seaborn-white')
plt.hist(H, bins='sturges',rwidth=0.99, color='b',density=True)
plt.xlabel('Height')
plt.ylabel('Frequency')
plt.title('Histogram')
plt.show()
```



Python Code

```
#====Histogram===#
#plt.savefig("D:/HIST.jpg", dpi=500)
plt.style.use('seaborn-white')
plt.hist(W, bins='sturges',rwidth=0.99,
   color='b',density=True)
plt.xlabel('Weight')
plt.ylabel('Frequency')
plt.title('Histogram')
plt.show()
```



Binomial Distribution

Exercise:

What do you mean by Binomial variate? Suppose follows Binomial distribution with parameter n and p then show that,

(i) $E\left[\frac{X}{n} - p\right]^2 = \frac{pq}{n}$

(ii) $Cov\left[\frac{X}{n}, \frac{n-X}{n}\right] = -\frac{pq}{n}$



Data Mining Techniques

Seventy Percent

Thirty Percent



Thank You....

